

Banking API Integration – Enterprise Grade Documentation (Swagger + Sequence + Multi-Tenant)

1. Enterprise Architecture Overview

Layered Architecture

Client Apps (Mobile / LOS / LMS / CRM) → API Gateway (Security + Routing) → Middleware / ESB (Transformation + Orchestration) → Microservices Layer (Customer / Loan / Payment Services) → Core Banking System (CBS) → Database Layer

2. Detailed Sequence Flow (Bank-Grade)

Use Case: Customer Creation API

1. Client sends request (POST /customer)
 2. API Gateway validates:
 3. Token
 4. Rate limit
 5. Request forwarded to Middleware
 6. Middleware:
 7. Adds tenant context
 8. Transforms JSON → CBS format
 9. Microservice validates business rules
 10. CBS persists data
 11. Response returned via same path
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3. Multi-Tenant API Design (VERY IMPORTANT)

Tenant Identification

Each request must carry tenant context:

Headers:

- X-Tenant-ID: BOB
- X-Channel: LOS / MOBILE / BRANCH

Data Isolation

- Separate schema per tenant OR
- Shared DB with Tenant ID column

Example

```
{ "tenantId": "BIL", "customerId": "C123" }
```

4. Enterprise API Standards

Naming Convention

- /api/v1/customers
- /api/v1/loans

HTTP Methods

- GET → Fetch
 - POST → Create
 - PUT → Update
 - DELETE → Remove
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5. Swagger / OpenAPI Specification (Machine Readable)

```
openapi: 3.0.0
info:
  title: Banking API
  version: 1.0.0
paths:
  /api/v1/customer/create:
    post:
      summary: Create Customer
      requestBody:
        required: true
        content:
          application/json:
            schema:
              type: object
```

```
      properties:
        customerId:
          type: string
        name:
          type: string
        dob:
          type: string
      responses:
        '200':
          description: Success
```

6. Detailed JSON Schema (Bank Grade)

Request Schema

```
{ "header": { "tenantId": "BIL", "channel": "LOS", "requestId": "UUID", "timestamp": "ISO8601" }, "body":
{ "customerId": "CUST123", "name": "John Doe", "dob": "1990-01-01", "kycStatus": "VERIFIED" } }
```

Response Schema

```
{ "header": { "status": "SUCCESS", "code": "200", "message": "Processed Successfully" }, "body":
{ "customerId": "CUST123" } }
```

7. Authentication – Enterprise Flow

OAuth 2.0 Flow

Step 1: Call Auth API POST /oauth/token

Step 2: Receive Token { "access_token": "abc123", "expires_in": 3600 }

Step 3: Use Token Authorization: Bearer abc123

8. API Gateway Policies

- Rate Limit: 1000 req/min
- IP Whitelisting
- JWT Validation
- Threat Protection

9. Middleware / ESB Logic

Responsibilities

- Routing
- Data Mapping
- Orchestration

Example Transformation

JSON → XML for CBS

10. Core Banking Integration Mapping

API Field	CBS Field
customerId	CIF_NO
name	CUSTOMER_NAME
dob	DATE_OF_BIRTH

11. Error Handling – Enterprise Standard

```
{ "error": { "code": "CBS_001", "message": "Customer already exists", "severity": "HIGH" } }
```

12. Async Processing (Advanced Banking)

When Used

- Payments
- Loan disbursement

Flow

API → Queue (Kafka) → Consumer → CBS

13. Logging & Audit (Regulatory Requirement)

- Request Log
- Response Log
- User Activity

- Audit Trail
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14. Performance Optimization

- Load Balancer
 - Caching (Redis)
 - Horizontal Scaling
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15. AI-Powered API Enhancements

- Smart Retry Mechanism
 - Fraud Detection APIs
 - Predictive Failure Alerts
 - Intelligent Routing
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16. End-to-End Example (Loan API)

POST /api/v1/loan/disburse

Flow: 1. LOS sends request 2. API Gateway validates 3. Middleware enriches data 4. CBS checks eligibility 5. Amount disbursed 6. Response returned

17. Testing Strategy (Bank Grade)

- Unit Testing
 - SIT (System Integration Testing)
 - UAT
 - Performance Testing (JMeter)
-

18. Deployment Architecture

- Dev → SIT → UAT → PROD
 - CI/CD Pipeline
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19. Security Compliance

- TLS 1.2+
- AES Encryption

- GDPR / PCI DSS Compliance
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20. Final Notes

This document is now aligned for:

- Bank RFP submissions
 - Client presentations
 - Technical implementation
 - SaaS multi-tenant platforms
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21. Next Level Upgrade (Optional)

- Add Postman Collection
- Add Real CBS Vendor Mapping (Temenos / Finacle)
- Add UI Screens for API Monitoring Dashboard